

the patients react and help them conquer their mind blocks.

Globally, doctors are experimenting with a concept called digital twinning. Research firm CBInsights defines, “A digital twin is a virtual copy of a physical entity or process, such as a patient, anatomical structure, or hospital environment. It’s modelled on data sources including electronic health records (EHRs), disease registries, wearable sensors, and more.” In telemedicine, digital twins would help doctors diagnose the problem better and provide a better prognosis. The doctors can trial the course of treatment on a digital twin before administering to the real patient. Powered by cloud data and AI, it also enables doctors to quickly compare the patients with similar cases, to treat them better. It can also potentially explain some grey areas like why certain patients respond better to a treatment than others.

Apart from profiling the patient, digital or virtual twinning technology can also be used to emulate the processes in the healthcare organisation, to make it more efficient. Companies like IBM and Dassault Systèmes are into digital twin tech. According to a report by Accenture, 66% of healthcare executives expect their organisation’s investment in intelligent digital twins to increase over the next three years. Research by MarketsAndMarkets indicates that the digital twin market is expected to grow to \$35.8 billion by 2025, healthcare and defence being amongst the top adopters.

The metaverse, with blockchain technology as its backbone, can also transform the way medical records are maintained. The open and secure nature of the blockchain ensures the safety of an individual’s medical records. It does away with fragmentation and redundancy, and makes the record-keeping transparent and efficient. It also prevents lock-in. When you want to consult a specialist from across the world, you do not have to wait for your



Figure 1: The living space of LG's Omnipod concept car adapts to the user's needs using a meta-environment screen (Courtesy: LG)

current physician to give you a copy of your records. You can retain control over your records and give access to a trusted individual at the click of a button! Chronicled, Ever, and Patientory are some of the blockchain based solutions for medical data management. EMRChain and Plenum Data are some of the homegrown solutions.

Product design: Defy the laws of the physical world

The field of product design is in for a mega upgrade in the metaverse era. There are three aspects to this: improvement in the design process, creation of new products for the metaverse, and metaverse integration in traditional physical products.

When you design a product for the metaverse, you can let your imagination run wild, and can create products that can otherwise exist only in dreams! The metaverse does not follow the laws of physics, so you can create drastically different products and experiences. For example, you can create a house on Mars, shoes that make you fly, or a car that hops!

You can also integrate the metaverse into a real, physical object. For example, an immersive lounge chair, which creates a real-time view of a beach or

valley of your choice. In CES 2022, LG showcased a concept car or mobile cabin called the Omnipod, a fully autonomous vehicle with an onboard living space that adapts to the passenger's unique needs. It has an expansive tunnel screen called the meta-environment screen, which stretches from the floor to the ceiling, helping to recreate a chosen environment, be it your lounge, workspace, bedroom, or garden! The concept cabin is replete with a virtual assistant called Reah, who also adapts to the chosen environment. For a spot of entertainment, you can connect the Omnipod's tunnel and multimedia screens to get a feeling of virtually entering the movie you are watching! There are exterior cameras, which let you also keep an eye on the road.

The metaverse can also drastically upgrade the very process of designing something. You can collaborate with people across the world to create cross-functional objects. You can first create 3D prototypes and test them in a virtual environment to make sure they work as intended before going to the next stage of development. Even for something as simple as a new series of totes or sneakers, you can first launch the designs as NFTs and gauge the customer response, before physically